

Hardware Integration for an Integrated Modular Motor Drive Including Distributed Control

Adam Shea

T.M. Jahns

Wisconsin Electric Machines and Power Electronics Consortium (WEMPEC)

Electrical and Computer Engineering

University of Wisconsin - Madison

Madison, WI USA

amshea@wisc.edu jahns@engr.wisc.edu

Abstract—The Integrated Modular Motor Drive (IMMD) concept is a promising approach to the physical integration of electric machines and drives that is based on the use of pole-drive building blocks consisting of segmented stator poles with concentrated windings, each excited by a dedicated single-phase inverter and controller. This paper presents details of the design, construction, and testing of an 18kW (peak) 6-phase IMMD demonstrator unit intended for a traction drive application using a surface PM machine. Key new features in this demo unit include the introduction of a dedicated controller in each of the drive units that shares responsibility for controlling the 6 phase-drive units using a heterarchical distributed control strategy. Experimental tests gathered to date have yielded promising results while also highlighting remaining challenges to full implementation of the IMMD concept.

I. INTRODUCTION

The increasing proliferation and maturation of electric drive technologies has caused drive system-level integration to become a high priority in a variety of application areas including electric propulsion, aerospace, white goods, and down-hole drilling. Tight integration of motor and drive electronics offers attractive properties including reduced overall system volume, reduction of high-current cabling, reduced radiated EMI, simplified cooling arrangements, and appealing fault tolerance opportunities [1-5].

However, physical integration of electric machines and drives presents many challenges. In order to integrate the controls and power electronics into the machine housing, the power electronics must be designed to operate reliably in elevated vibration and thermal environments that fall outside of standard ranges for industrial-grade components[6]. Further complicating the challenges, customers for integrated drives typically expect the drive electronics to meet or exceed the lifetime of the machine, requiring very high reliability in demanding environments.

Early generations of integrated motor drives have typically used conventional three-phase induction or PM synchronous machines combined with standard voltage-source 6-switch bridge inverters to excite the machine. In many of these units, the drive is housed in its own enclosure which, in turn, is mounted on the side or end of the machine

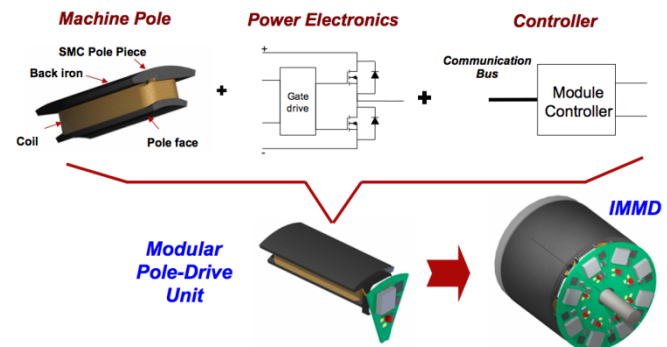


Figure 1: Integrated Modular Motor Drive (IMMD) concept

[7]. More aggressive concepts for integrated machine drives have been proposed that approach more closely the ultimate objective of mounting the complete drive inside the same enclosure as the machine [1],[4], but commercialization of these advanced integrated drive architectures has been limited to date.

One of these advanced concepts that has been proposed in the literature is the integrated modular motor drive (IMMD) [2]. As illustrated in Fig. 1, this concept segments the stator into individual pole pieces, each encircled by a concentrated winding that is excited by its own dedicated electronics unit. Each dedicated unit includes both the necessary power electronics and controls to form a complete pole-drive unit. The IMMD is completed by interconnecting a number of these pole-drive units to form an annulus around the rotor. Appealing features of this IMMD concept emerge from its modularity, offering opportunities for improved manufacturability, enhanced fault tolerance and redundancy, and simplified repair/replacement of the pole-drive units.

Challenges associated with the IMMD configuration include the demanding thermal, vibrational, and EMI environments in close proximity to the machine, and higher parts count that influences reliability and cost. There are additional challenges introduced when higher levels of fault tolerance are desired, and some of those challenges will be discussed in this paper.

Previous work on the IMMD concept has led to the fabrication of concept demonstration units that incorporate

the segmentation of the power electronics but retained a single centralized controller operating all of the pole-drive units [2]. The purpose of this paper is to present the details of a new generation of IMMD demonstrator unit that segments the controller as well. More specifically, the hardware design of a 6-phase IMMD for a 10kW surface PM machine is presented in which each phase is independently controlled by its own processor packaged with the pole-drive unit.

Details of the resulting distributed controls architecture are discussed in order to provide insights into the balance struck between the need for cooperation among the individual controllers and the desire for maximizing the independence of each controller for purposes of reliability and fault tolerance. The goal is to highlight the design features that have been introduced to develop an integrated design while addressing the competing demands of the power electronics, controllers, and sensors that influence power circuit parasitic components, thermal management, EMC characteristics, and, ultimately, cost. Finally, test results from the IMMD hardware demonstrator unit are presented to highlight the progress achieved to date as well as the directions of future work.

II. IMMD MACHINE AND CONTROLS TOPOLOGY

A. IMMD Machine and Drive Configuration

Although the focus of this paper is the IMMD controller, it is important to properly introduce the chosen machine and drive configuration that establishes the framework for the controller development. Since a more detailed discussion of the IMMD machine topology study is available in a separate paper [12], the discussion here will be limited to a summary of the final selection and the associated rationale.

After completing a thorough evaluation process of candidate IMMD machine topologies, the decision was made to select a six-phase variant of a three-phase fractional-slot concentrated winding permanent magnet (PM) synchronous machine. Inspection of the selected machine cross section shows that the stator has 12 slots, combined with a 10-pole surface-mounted PM rotor. The figure shows that the stator is partitioned into six identical arcs, each consisting of two adjacent stator teeth with a concentrated coil surrounding each tooth. This arc structure with its two windings connected in either series (typical) or parallel comprises the basic machine pole unit for one phase of the IMMD. The figure shows that each of these 6 pole units occupies a 60 degree arc along the stator periphery when they are combined to form the annular stator structure.

The major factors favoring this six-phase machine for the IMMD over the other candidates include: 1) its lower machine mass compared to its major competitors; its lower fundamental frequency at maximum speed (1.16 kHz @ 14,000 r/min) that is critical to achieving high efficiency values at elevated speeds; and 3) its symmetry that

eliminates unbalanced magnetic pull on the rotor during balanced 6-phase excitation. Although its winding factor is not the highest among the candidate machines, the overall evaluation of this configuration gave it a net advantage over the other alternative machine configurations that ultimately led to its selection as the preferred topology for further development.

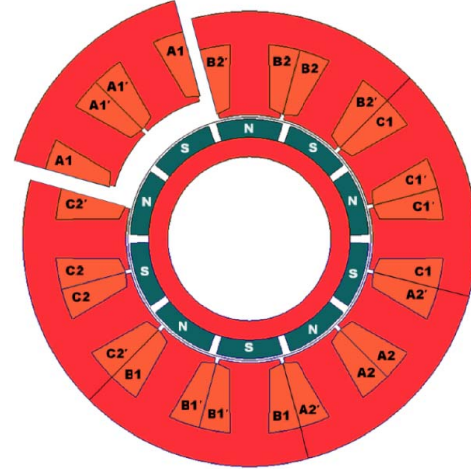


Figure 2: 12-slot/10-pole FSCW PM machine with segmented stator and 6 stator phase selected for IMMD demonstrator unit.

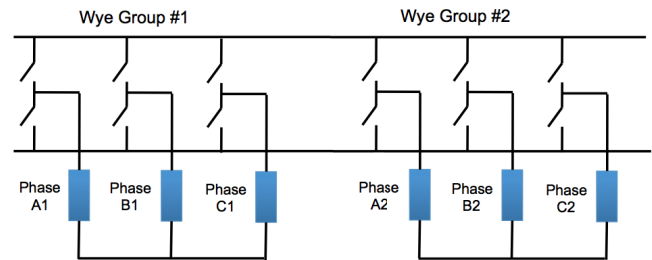


Figure 3: Basic 6-phase IMMD demonstrator drive configuration with two three-phase floating wye subcircuits and shared dc bus

The choice of the surface PM rotor for the selected machine in Fig. 2 was convenient for this project but it is not exclusive. That is, the surface PM rotor can be readily replaced with an interior PM alternative, and the resulting stator-rotor combination works very well. Unfortunately, the alternative choice of an induction machine rotor will not work well because of the rich spectrum of spatial harmonic flux density components that appear in the airgap of the FSCW stator. These harmonics generate parasitic losses in the cage rotor that are unacceptably high for most applications.

There are also several alternative inverter configurations for exciting a six-phase machine. As was the case for the machine, the advantages and limitations of the major candidates were evaluated and compared, leading to the choice of the double floating wye configuration shown in

Fig. 3. Although there are alternative inverter schemes with higher fault tolerance if a four-switch H-bridge is assigned to each phase, the double wye has the advantage of only two switches per phase. In addition, the presence of two wyes makes it possible to shut one down one three-phase inverter and continue operating with half of the machine excited under fault conditions. While Fig. 3 shows a shared dc bus for both inverters, the natural fault tolerance of the configuration can be increased by breaking the bus into two halves with an independent dc source for each of the wyes.

A more detailed schematic of the power circuit schematic for the double-wye IMMD configuration is provided in Fig. 4. This circuit makes it clearer that each of the 6 stator double-tooth pole units identified in Fig. 3 is excited with its own dedicated half-bridge inverter. These single-phase inverter units are shown as shaded boxes in the figure. The circuit diagram also indicates that the dc link inverter capacitance is broken into six equal sections and distributed among the six phase-drive units.

Another feature in the Fig. 3 power circuit diagram that can be noted is the interspersing of phases from the two wyes in an alternating sequence around the periphery of the stator. During normal operation, each pair of pole-units on opposite sides of the stator are from the same phase group (e.g., A1 and A2) so they are excited with the same synchronized waveform (except for a polarity reversal). Under these conditions, the alternating sequence of the phases does not matter. However, if excitation is removed from one of the wyes, the remaining three excited phases are distributed evenly around the stator's periphery, minimizing unbalanced radial forces on the rotor that can stress the rotor bearings.

In order to achieve tight packaging for the machine and drive, the total amount of dc bus capacitance must be minimized. Accomplishing this objective is complicated by the adopted restriction that electrolytic capacitors cannot be used due to the high-temperature environment. To help cope with this limitation, the dc link was configured as a ring-star interconnection consisting of individual star connections to each dc-link power input and a ring of connections between adjacent modules to share ripple current. This configuration offers advantages for canceling switching harmonics. It physically distributes the dc link capacitor between phase modules as indicated in Fig. 4, but also splits it into three different types of capacitors (film, ceramic, and planar bus) that act to provide a consistently low impedance over a wide range of frequencies [2],[11].

B. IMMD Controller Configuration

In order to achieve higher levels of fault tolerance, single-point failures must be eliminated from the drive system to the greatest possible extent. In conventional drive topologies, the centralized controller is one of the major sources of vulnerability to single-point failures. As noted in the introduction, each pole-drive unit in the IMMD

configuration, constituting one of the drive phases, is controlled by its own processor.



Figure 4: Power schematic of 6-phase modular motor drive showing dc-link, paralleled decoupling capacitors, power switches, and two machine star-points.

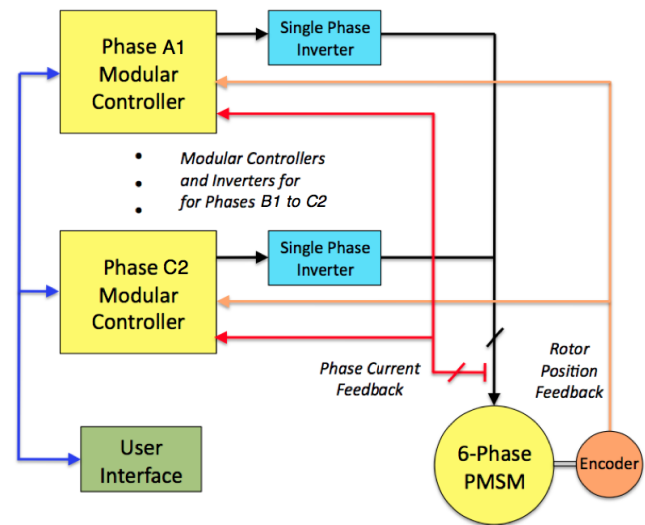


Figure 5: Diagram of basic IMMD distributed control configuration customized for 6-phase motor topology

There are several alternative control algorithms that have been proposed and developed for distributed control architectures with multiple processors. One of the more popular alternatives is a master/slave control configuration, in which one of n distributed controllers is designated to act as the master controller at every time instant.

However, the distributed control algorithm that was found to be most suitable for the 6-phase IMMD is heterarchical control [8] in which the six distributed controllers operate as synchronized peers to run identical control programs while sharing sensor information as needed (see Fig. 5). For the IMMD, this sharing of sensor feedback includes rotor position feedback from an encoder that is delivered to all 6 controllers, and phase current feedback that is shared between adjacent phases. This approach strikes a

compromise that makes it possible to achieve acceptable machine control performance during normal operation with a minimal amount of required communications bandwidth [9-10] between the phase-unit processors.

III. IMMD DEMONSTRATOR HARDWARE DESIGN

A. IMMD Demonstrator Unit Ratings

The basic IMMD concept is flexible and scalable, making this integrated motor/controller configuration a candidate for a wide range of power ratings and applications. However, this particular project has been directed to electric traction applications, which is one of the major areas where automotive manufacturers have expressed significant interest in approaches for combining the machine and power controller into the same structure [13].

The advanced traction drive requirements that have provided the inspiration for the target power and performance requirements adopted for the IMMD demonstrator unit originated with the FreedomCar (now U.S. DRIVE) program sponsored by the US Department of Energy in collaboration with several major automobile manufacturers in the United States. Although the complete set of requirements for the advanced traction motor and drive defined by the U.S. Drive program will not be repeated here [14], a few key performance targets will be highlighted. In particular, the traction motor is expected to deliver 30 kW continuously and 55 kW for 18 s at a corner-point speed of 2800 r/min, with 30 kW of constant power expected between 2800 and 14,000 r/min, the maximum speed.

For this concept demonstrator project, the power requirement were scaled down by a factor of 3 to 10 kW continuous and 18 kW (peak) at the unchanged corner-point speed of 2800 r/min. This corresponds to continuous and peak torque requirements of 34.1 and 61.4 Nm, respectively.

While the U.S. DRIVE requirements call for operation of the inverter using coolant at 105°C, these thermal requirements were relaxed for the demonstrator IMMD in order to make it possible to use off-the-shelf industry-grade components with silicon-based semiconductors. However, some initial steps were taken in the direction of high-temperature operation, including the elimination of electrolytic capacitors in favor of higher-temperature film and ceramic capacitors. Other steps taken toward the implementation of a high-temperature controller that is capable of meeting the 105°C coolant requirement are discussed later in this section.

B. IMMD Demonstrator Hardware Design Issues

1) Power Converter and Controller Assembly

The controller hardware for this IMMD demonstrator unit has been designed using printed circuit board (PCB) modules with bolt interconnections for each phase that allows for easy assembly and quick replacement in case of failures. The key dimensions of the phase-drive PCB have been chosen to match those of the adjacent machine segment

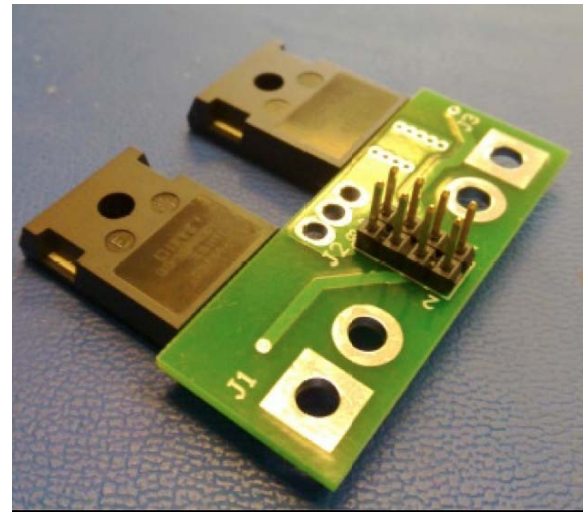


Figure 6: View of half-bridge power switch module.

(i.e., 1/6th of the stator circumference) that the phase-drive unit excites. A stainless-steel backplate is mounted between the machine and the drive electronics to shield the electronics from fringing magnetic and electrical fields radiated by the machine.

Since this demonstrator is only designed to deliver a maximum peak power of 18kW, the half-bridge inverter phase-leg for each phase has been designed to use two discrete 600V IGBTs in TO-247 packages rather than using a half-bridge power module that would likely be the preferred choice for higher-power designs. The phase-leg power stage shown in Fig. 6 contains the two IGBTs for a single phase along with screw terminals for the dc link connections, an RTC temperature sensor, gate connections, and a mounting terminal for the machine phase busbar. The nominal dc bus voltage is 325 Vdc.

Cooling for each phase-leg module is accomplished using an off-the-shelf copper pin-fin waterblock designed for graphics card cooling. As shown in Fig.7 and Fig. 8, the power module is mounted in direct contact with the waterblock to minimize the thermal resistance.

The control board is mounted above the power module and waterblock. This control board is designed with 4 layers of copper traces. The bottom two are solid planes that form the dc-link buswork for the phase-drive unit while also creating a very low-inductance 1nF capacitor. The top two layers contain the DSP controller, analog signal conditioning for voltage and current sensing, gate drivers, and communications interfaces.

The dc-link bulk capacitance is a 20μF film capacitor mounted below the controller on each module. Middle-range frequencies are decoupled using a pair of 22nF ceramic capacitors mounted very near the dc link terminals of each power module. All of these discrete capacitors are identified in the lower half of Fig. 7. This distributed capacitance

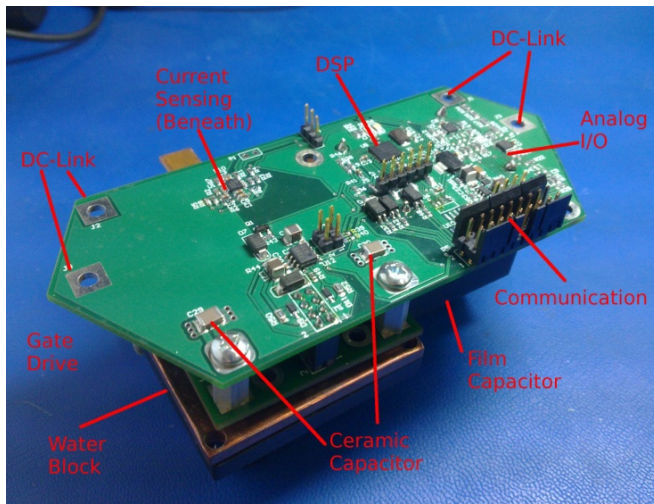


Figure 7: Top view of complete drive-phase module with labeled components.

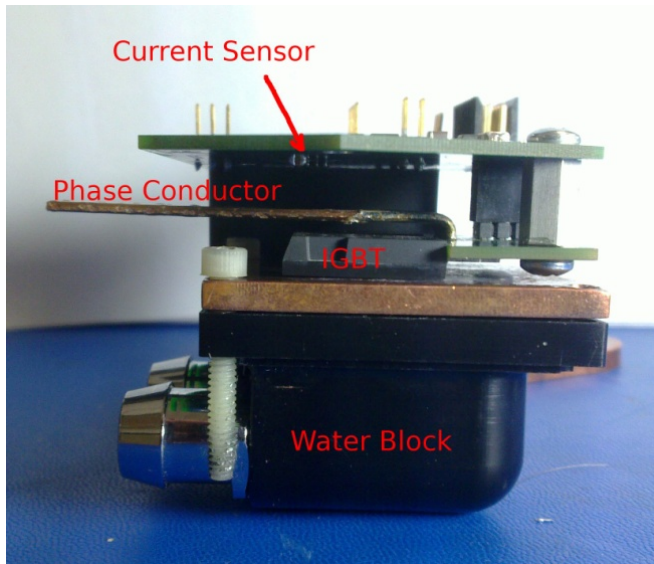


Figure 8: Side view of complete drive-phase module with labeled components.

makes it possible to reduce the total capacitance needed to meet the low-impedance requirements of the inverter dc link.

The gate drive consists of a simple bootstrap driver for the high-side IGBT switch with a matching power driver for the low-side switch. A voltage-clamped version of the half-bridge output node voltage is fed back to the controller in order to provide the necessary feedback for IGBT desaturation protection.

2) Sensors

High-bandwidth ($\sim 10\text{kHz}$) current sensing is provided by a linear Hall effect sensor mounted to the bottom of the control board. This sensor measures the field produced by current flowing in the machine phase conductor mounted directly below it. The sensor is shielded from the machine by both the stainless-steel baseplate for low frequencies and

by the copper face of the waterblock for higher frequencies. The mounting standoffs control the spacing in order to permit easy sensor calibration.

The Hall sensor signal is buffered and sent to both the controller and the two adjacent phase modules, one in the same machine wye, and one on the companion wye (see Fig. 5) in order to provide accurate current vector feedback information and to make fast fault detection and response possible.

Position sensing in this demonstrator unit is accomplished using a standard quadrature encoder. The angular position information is delivered in parallel to all six phase-drive controllers. It is recognized and acknowledged that this position sensor represents one of the key remaining vulnerabilities to single-point failures. Development of approaches for mitigating or eliminating this vulnerability is a topic of continuing research that is beyond the scope of this paper.

3) Communications and Grounding

Module-to-module communications are handled by both analog signals for current sensing and a CAN bus. The CAN bus, encoder interface, and controls power supply are isolated from the power circuit by means of an isolated dc/dc converter in each module and optical isolation for the signal lines. For grounding purposes, the controls are all connected to the negative rail of the dc bus in each module, providing a single common reference point. Potential ground loops in the current feedback signals are decoupled using differential signaling.

4) Thermal Management

In order to allow coolant to be shared between the machine and drive, IGBTs rated for a maximum junction temperature of 175°C were chosen along with capacitors rated at 125°C and a high-temperature DSP controller rated at 140°C . Simulations of full-power operation predict a 5°C margin when operating with 100°C coolant for these components.

Unfortunately, the complete phase-drive module cannot be expected to operate at these elevated temperatures since some of the other components including the bootstrap gate driver and opto-isolators could not be procured with sufficiently high temperature ratings.

IV. EXPERIMENTAL TEST RESULTS

The complete IMMD electronics assembly consisting of 6 interconnected phase-drive units has been mounted on a dynamometer in close proximity to a 6-phase, 18kW (peak) SPM machine (12 slots, 10 poles) that was built for this IMMD project [12]. Fig. 9 provides a view of the combined IMMD demonstrator unit mounted on the dynamometer.

To date, preliminary testing has been carried out to debug the electronics and to begin exploring the performance

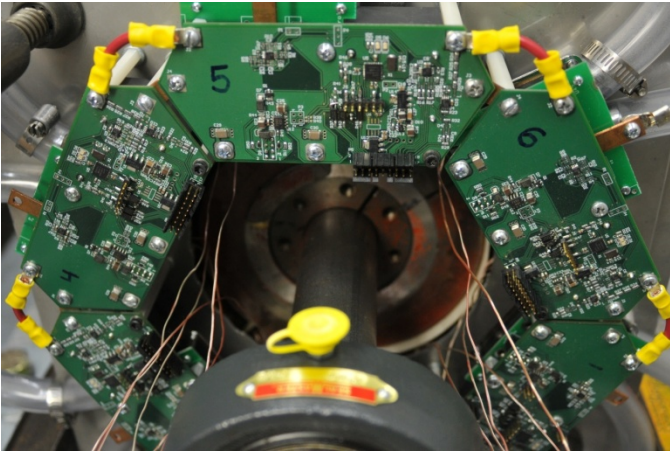


Figure 9: 6-phase IMMD demonstrator unit mounted on dynamometer with test machine.

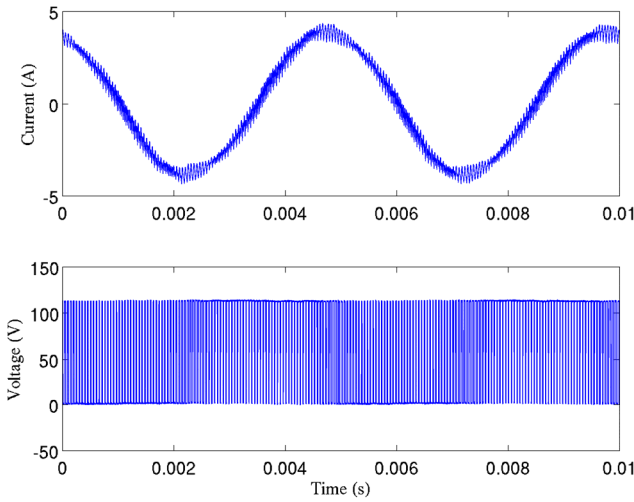


Figure 10: Demonstrator IMMD phase current and PWM voltage waveforms for constant V/Hz operation at 2400 r/min, $V_{dc}=110$ V.

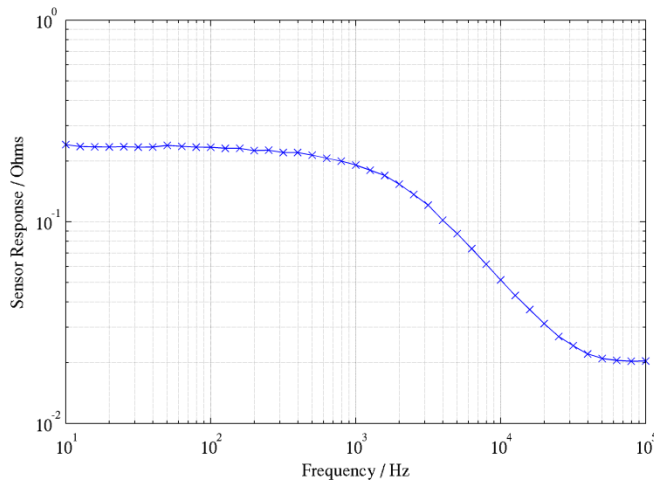


Figure 11: Hall-effect magnetic field-based current sensor frequency response.

characteristics of the IMMD demonstrator unit. Phase current and voltage waveforms for one of the six phases during constant-V/Hz operation at 200Hz (2400r/min) under no-load conditions are provided in Fig. 10. A basic sine-triangle PWM algorithm is used for the voltage modulation with a switching frequency of 20kHz and a modulation index value of 0.95. For this preliminary test, the drive is operating with a reduced dc link voltage of 110V, approx. one-third the rated dc link voltage of 325V. The quality of the phase current waveform is quite promising.

The current sensing scheme using the Hall magnetic field sensor has been tested. The large distance between phase conductors and the shielding effect of the copper cooling plates has succeeded in suppressing the inter-phase cross-coupling below the detection threshold. The frequency response of the system is shown in Fig. 11. The frequency cutoff was chosen by filter design to fall just below the PWM switching frequency in order to enable good control loop response of the phase current regulators in the presence of non-synchronized PWM carrier waveforms.

V. CONCLUSIONS

The IMMD demonstrator unit discussed in this paper represents a significant step towards full implementation of the IMMD concept. More specifically, the incorporation of separate controllers for each of the six phase-drive units in this demonstrator unit has provided a valuable opportunity to address the challenges associated with distributed control of the machine phases. Each phase in the demonstrator unit is equipped with its own controller, gate drive, and current sensor packaged in a compact and easily-replaceable unit.

Experimental results gathered to date indicate that the distributed control configuration adopted for this demo unit is capable of delivering the desired performance characteristics. Continuing work on the IMMD concept is focused on more detailed investigations of the distributed control strategies under both normal and faulted operating conditions. Techniques for eliminating the remaining IMMD vulnerabilities to single-point failure are targeted for future investigation.

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